Tyrel Stott

ISIN 325

Dr. Gogolin

April 5, 2026

SQL 3

This is my SQL 3 assignment for ISIN 325. In this we utilize some basic coding techniques like if statements and some loops. We also employ some more SQL techniques such as truncating tables and start to implement logging. I found the logging portion of this assignment to be quite difficult. I created the table within the database to hold log information like we did in class, but when I use it for any other script besides my first, it won't add any information. Additionally, it wouldn't recognize the database as existing despite me verifying it is there correctly. I am a bit baffled by this. Despite this, I did include the logging routine and do understand how it works.

You're running Visual Studio as a different user than the one logged into Windows. Depending on your configuration, you may experience a degraded sign-in experience for bo

Object Explorer

Connect

- localhost\SQLEXPRESS02 (SQL Server 17.0.1000 - TYRELS-L
  - Databases
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    - Meijer
      - Database Diagrams
      - Tables
        - System Tables
        - FileTables
        - External Tables
        - Graph Tables
        - dbo.Category
        - dbo.Customer
        - dbo.Customer2
          - Columns
            - CustomerID (int, not null)
            - FirstName (varchar(50), null)
            - LastName (varchar(50), null)
            - DateOfBirth (date, null)
            - Address (varchar(100), null)
            - City (varchar(50), null)
            - State (char(2), null)
            - ZipCode (varchar(10), null)
          - Keys
          - Constraints
          - Triggers
          - Indexes
          - Statistics
        - dbo.Product
        - dbo.Purchase
        - dbo.PurchaseDetail
        - dbo.Store
        - dbo.StoreProduct
        - dbo.Supplier
      - Views
      - External Resources
      - Synonyms
      - Programmability
      - Query Store
      - Service Broker
      - Storage
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    - Sick
    - Sick2
  - Security
  - Server Objects
  - Replication
  - Management

SQLQuery2.s...IADMIN (89))* | SQLQuery1....ADMIN (80))*

```sql
Use Meijer


-- Question 1, use the normal SQL to define the table...
CREATE TABLE Customer2 (
    CustomerID INT IDENTITY(100,1),
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DateOfBirth DATE,
    Address VARCHAR(100),
    City VARCHAR(50),
    State CHAR(2),
    ZipCode VARCHAR(10)
);
```

107 % No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-05T11:48:32.8077083-04:00
```

107 % No issues found

Query executed successfully.

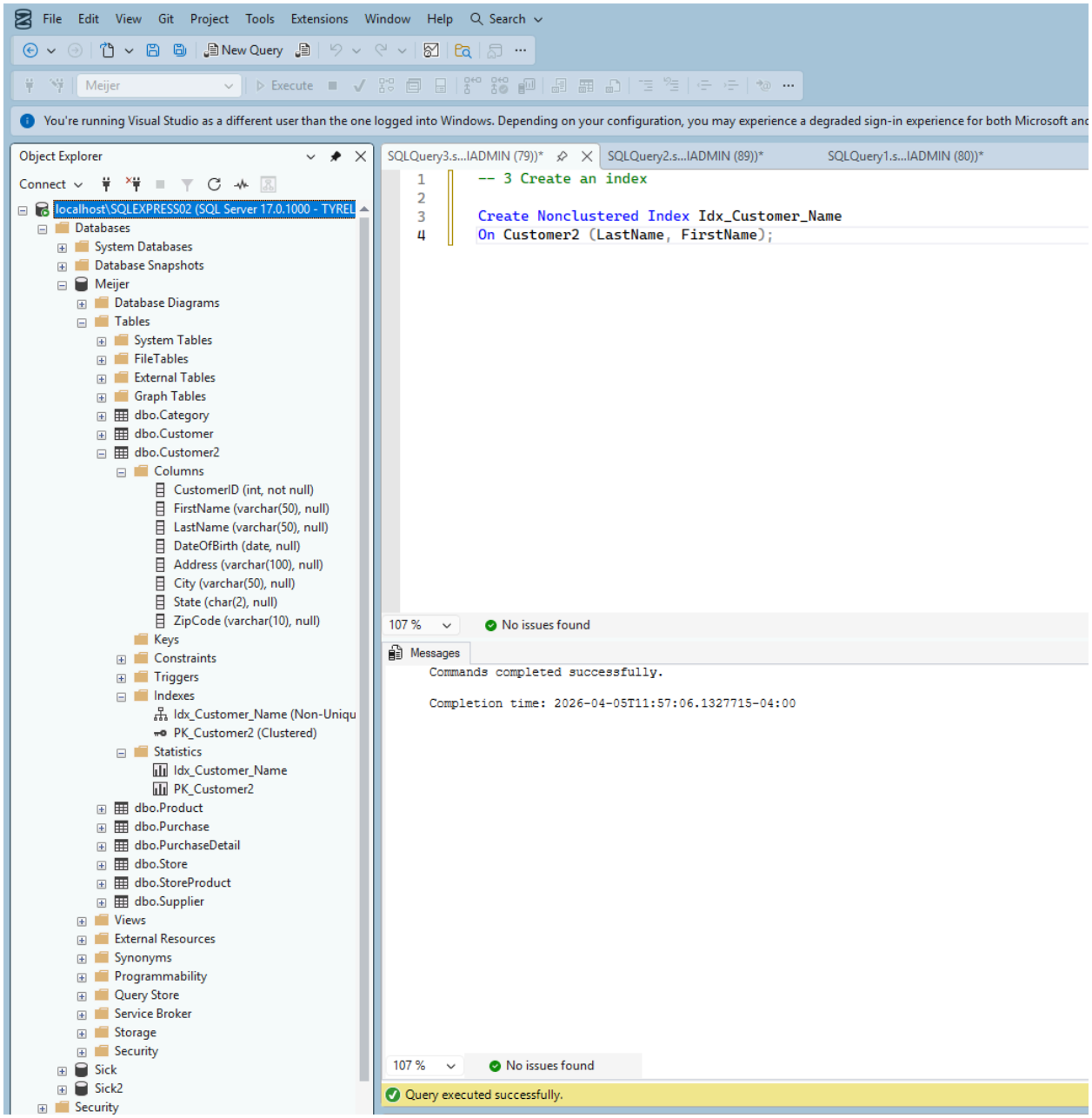
SQLQuery2....ADMIN (89))* | SQLQuery1.s...IADMIN (80))*

```sql
-- 2 Alter the table..

Alter Table Customer2

Add constraint PK_Customer2 Primary Key (CustomerID);
```

107 % | No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-05T11:50:39.1427888-04:00
```


File Edit View Git Project Tools Extensions Window Help Search
New Query
Meijer
Execute
You're running Visual Studio as a different user than the one logged into Windows. Depending on your configuration, you may experience a degraded sign-in experience for both Microsoft and
Object Explorer
Connect
localhost\SQLEXPRESS02 (SQL Server 17.0.1000 - TYREL
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System Databases
Database Snapshots
Meijer
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.Category
dbo.Customer
dbo.Customer2
Columns
CustomerID (int, not null)
FirstName (varchar(50), null)
LastName (varchar(50), null)
DateOfBirth (date, null)
Address (varchar(100), null)
City (varchar(50), null)
State (char(2), null)
ZipCode (varchar(10), null)
Keys
Constraints
Triggers
Indexes
Idx_Customer_Name (Non-Uniqu
PK_Customer2 (Clustered)
Statistics
Idx_Customer_Name
PK_Customer2
dbo.Product
dbo.Purchase
dbo.PurchaseDetail
dbo.Store
dbo.StoreProduct
dbo.Supplier
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security
Sick
Sick2
Security
SQLQuery3.s...IADMIN (79))*
SQLQuery2.s...IADMIN (89))*
SQLQuery1.s...IADMIN (80))*
-- 3 Create an index
Create Nonclustered Index Idx_Customer_Name
On Customer2 (LastName, FirstName);
107 %
No issues found
Messages
Commands completed successfully.
Completion time: 2026-04-05T11:57:06.1327715-04:00
107 %
No issues found
Query executed successfully.

Meijer | Execute

You're running Visual Studio as a different user than the one logged into Windows. Depending on your configuration, you may experience a degraded sign-in experience for both Microsoft and GitHub accounts. Dismiss Don't show a

Object Explorer

Connect

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            - City (varchar(50), null)
            - State (char(2), null)
            - ZipCode (varchar(10), null)
          - Keys
          - Constraints
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          - Indexes
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        - dbo.Product
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      - Service Broker
      - Storage
      - Security
    - Sick
    - Sick2
  - Security

SQLQuery4....ADMIN (67))* | SQLQuery3.s...IADMIN (79))* | SQLQuery2.s...IADMIN (89))* | SQLQuery1.s...IADMIN (80))*

```sql
-- Insert 50 rows, used chatgpt to generate the data
Use Meijer

INSERT INTO Customer2 (FirstName, LastName, DateOfBirth, Address, City, State, ZipCode)
VALUES
('Liam','Anderson','1990-01-12','101 Maple St','Detroit','MI','48201'),
('Olivia','Brown','1988-03-22','202 Oak St','Lansing','MI','48901'),
('Noah','Clark','1992-05-14','303 Pine St','Flint','MI','48501'),
('Emma','Davis','1995-07-30','404 Elm St','Grand Rapids','MI','49501'),
('Oliver','Evans','1987-09-18','505 Cedar St','Ann Arbor','MI','48103'),
('Ava','Garcia','1991-11-25','606 Birch St','Kalamazoo','MI','49001'),
('Elijah','Harris','1989-02-10','707 Spruce St','Saginaw','MI','48601'),
('Sophia','Johnson','1993-04-05','808 Willow St','Muskegon','MI','49440'),
('James','King','1994-06-17','909 Cherry St','Midland','MI','48640'),
('Isabella','Lewis','1990-08-29','111 Walnut St','Bay City','MI','48706'),

('Benjamin','Martinez','1986-10-13','222 Poplar St','Traverse City','MI','49684'),
('Mia','Moore','1992-12-01','333 Ash St','Jackson','MI','49201'),
('Lucas','Nelson','1985-01-19','444 Beech St','Battle Creek','MI','49014'),
('Charlotte','Ortiz','1993-03-27','555 Hickory St','Port Huron','MI','48060'),
('Henry','Perez','1988-05-09','666 Dogwood St','Monroe','MI','48161'),
('Amelia','Roberts','1991-07-21','777 Sycamore St','Mount Pleasant','MI','48858'),
('Alexander','Scott','1994-09-11','888 Fir St','Ypsilanti','MI','48197'),
```

107 % | 8 | 0

Messages

```
(50 rows affected)

Completion time: 2026-04-05T11:59:41.2975568-04:00
```

107 % | No issues found

Query executed successfully.

SQLQuery5.s...ADMIN (72)) | SQLQuery4.s...IADMIN (67))* | SQLQuery3.s...IADMIN (79))* | SQLQu

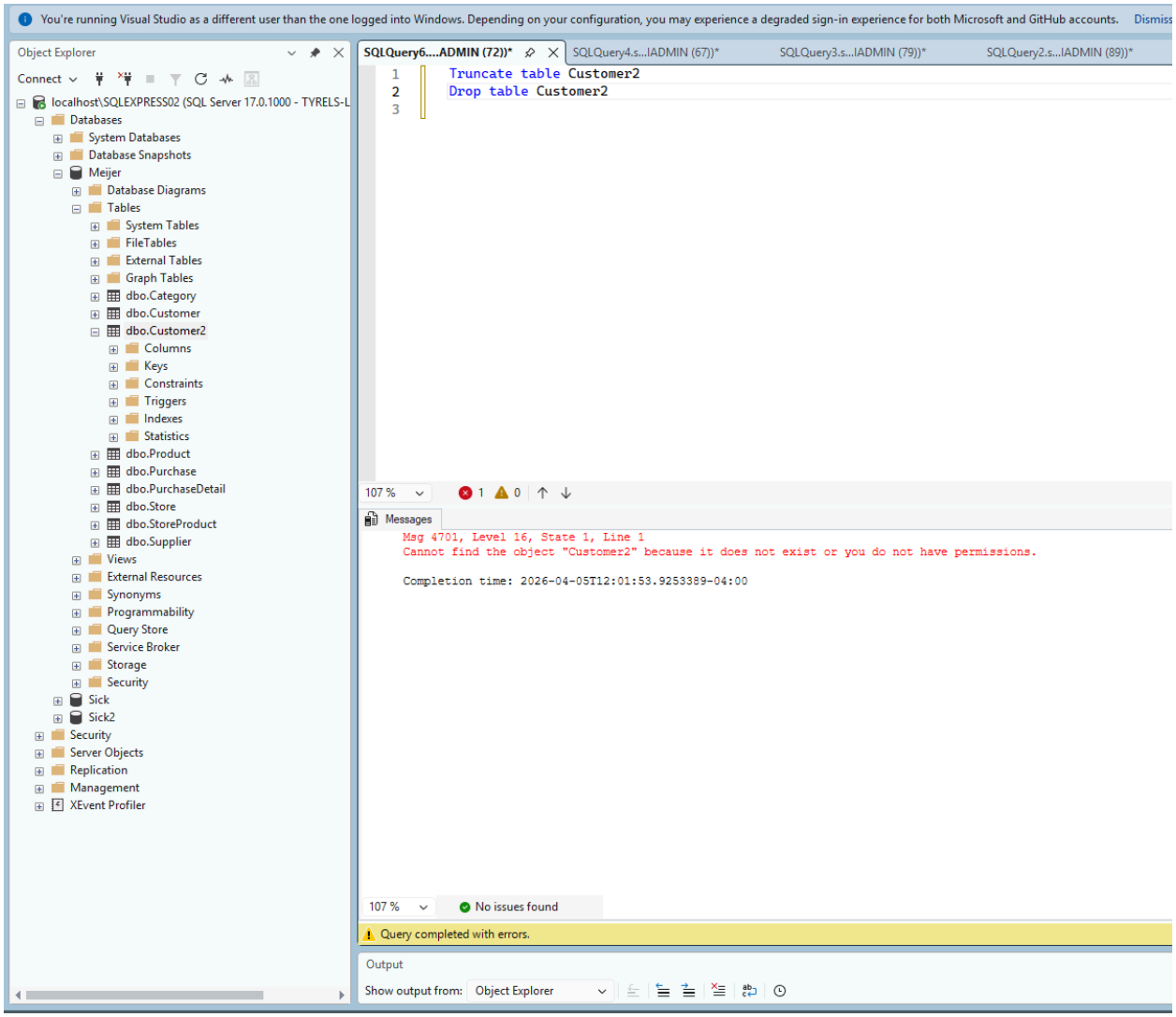
```
SELECT TOP (1000) [CustomerID]
        ,[FirstName]
        ,[LastName]
        ,[DateOfBirth]
```

107 % No issues found

Results | Messages

	CustomerID	FirstName	LastName	DateOfBirth	Address	City	State	ZipCode
1	100	Liam	Anderson	1990-01-12	101 Maple St	Detroit	MI	48201
2	101	Olivia	Brown	1988-03-22	202 Oak St	Lansing	MI	48901
3	102	Noah	Clark	1992-05-14	303 Pine St	Flint	MI	48501
4	103	Emma	Davis	1995-07-30	404 Elm St	Grand Rapids	MI	49501
5	104	Oliver	Evans	1987-09-18	505 Cedar St	Ann Arbor	MI	48103
6	105	Ava	Garcia	1991-11-25	606 Birch St	Kalamazoo	MI	49001
7	106	Elijah	Harris	1989-02-10	707 Spruce St	Saginaw	MI	48601
8	107	Sophia	Johnson	1993-04-05	808 Willow St	Muskegon	MI	49440
9	108	James	King	1994-06-17	909 Cherry St	Midland	MI	48640
10	109	Isabella	Lewis	1990-08-29	111 Walnut St	Bay City	MI	48706
11	110	Benjamin	Martinez	1986-10-13	222 Poplar St	Traverse City	MI	49684
12	111	Mia	Moore	1992-12-01	333 Ash St	Jackson	MI	49201
13	112	Lucas	Nelson	1985-01-19	444 Beech St	Battle Creek	MI	49014
14	113	Charlotte	Ortiz	1993-03-27	555 Hickory St	Port Huron	MI	48060
15	114	Henry	Perez	1988-05-09	666 Dogwood St	Monroe	MI	48161
16	115	Amelia	Roberts	1991-07-21	777 Sycamore St	Mount Pleasant	MI	48858
17	116	Alexander	Scott	1994-09-11	888 Fir St	Ypsilanti	MI	48197
18	117	Harper	Turner	1987-11-03	999 Palm St	Adrian	MI	49221
19	118	Daniel	Walker	1990-02-14	121 Lake St	Marquette	MI	49855
20	119	Evelyn	Young	1995-04-26	232 Hill St	Escanaba	MI	49829
21	120	Matthew	Allen	1989-06-08	343 River St	Alpena	MI	49707
22	121	Abigail	Baker	1993-08-19	454 Forest St	Petoskey	MI	49770
23	122	Joseph	Campbell	1986-10-30	565 Valley St	Cadillac	MI	49601
24	123	Emily	Carter	1991-12-12	676 Sunset St	Big Rapids	MI	49307
25	124	David	Collins	1988-01-24	787 Sunrise St	Gaylord	MI	49735
26	125	Elizabeth	Edwards	1994-03-06	898 Meadow St	Holland	MI	49423
27	126	Michael	Flores	1987-05-18	909 Brook St	Zeeland	MI	49464
28	127	Sofia	Gonzalez	1992-07-29	101 Creek St	Niles	MI	49120
29	128	William	Green	1990-09-10	212 Ridge St	Sturgis	MI	49091
30	129	Avery	Hall	1995-11-22	323 Prairie St	Coldwater	MI	49036
31	130	Ethan	Hill	1989-02-03	434 Grove St	Owosso	MI	48867
32	131	Ella	Howard	1993-04-15	545 Park St	Corunna	MI	48817
33	132	Jacob	Jenkins	1986-06-27	656 Center St	Charlotte	MI	48813
34	133	Scarlett	Kelly	1991-08-08	767 Front St	Ionia	MI	48846
35	134	Logan	Long	1988-10-19	878 Market St	Greenville	MI	48838
36	135	Grace	Mitchell	1994-12-01	989 Water St	Lowell	MI	49331
37	136	Jackson	Morgan	1987-01-13	111 Bridge St	Rockford	MI	49341
38	137	Chloe	Morris	1992-03-25	222 Mill St	Sparta	MI	49345
39	138	Levi	Murphy	1990-05-07	333 Depot St	Belding	MI	48809

Query executed successfully.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Server Objects
Replication
Management
XEvent Profiler
SQLQuery6....ADMIN (72))*
SQLQuery4.s...IADMIN (67))*
SQLQuery3.s...IADMIN (79))*
SQLQuery2.s...IADMIN (89))*
1 Truncate table Customer2
2 Drop table Customer2
3
107 %
1
0
Messages
Msg 4701, Level 16, State 1, Line 1
Cannot find the object "Customer2" because it does not exist or you do not have permissions.
Completion time: 2026-04-05T12:01:53.9253389-04:00
107 %
No issues found
Query completed with errors.
Output
Show output from: Object Explorer

SQLQuery7....ADMIN (52))* | SQLQuery6.s...lADMIN (72))* | SQLQuery4.s...lADMIN (67))* | SQLQuery3.s...lADMIN (79))* | SQLQ

```sql
CREATE VIEW vw_Customer2
AS
SELECT CustomerID, FirstName, LastName, Address, City, State, ZipCode
FROM Customer2;

-- 6 Create view, I repopulated the customer2 database using the other sql for this one.
```

107 % | No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-05T12:09:48.1186531-04:00
```

SQLQuery8....ADMIN (75))* | SQLQuery7.s...lADMIN (52))* | SQLQuery6.s...lADMIN (72))* | SQLQuery4.s...lADMIN (67))*

```sql
select *
from vw_Customer2
Order by FirstName DESC;
```

107 % 2 0

Results | Messages

	CustomerID	FirstName	LastName	Address	City	State	ZipCode
1	147	Zoey	Watson	343 Jefferson St	South Haven	MI	49090
2	128	William	Green	212 Ridge St	Sturgis	MI	49091
3	139	Victoria	Parker	444 Main St	Hastings	MI	49058
4	107	Sophia	Johnson	808 Willow St	Muskegon	MI	49440
5	127	Sofia	Gonzalez	101 Creek St	Niles	MI	49120
6	140	Sebastian	Price	555 Union St	Marshall	MI	49068
7	133	Scarlett	Kelly	767 Front St	Ionia	MI	48846
8	142	Owen	Rogers	777 College St	Springfield	MI	49037
9	101	Olivia	Brown	202 Oak St	Lansing	MI	48901
10	104	Oliver	Evans	505 Cedar St	Ann Arbor	MI	48103
11	149	Nora	Wright	565 Jackson St	Manistee	MI	49660
12	102	Noah	Clark	303 Pine St	Flint	MI	48501
13	126	Michael	Flores	909 Brook St	Zeeland	MI	49464
14	111	Mia	Moore	333 Ash St	Jackson	MI	49201
15	120	Matthew	Allen	343 River St	Alpena	MI	49707
16	112	Lucas	Nelson	444 Beech St	Battle Creek	MI	49014
17	134	Logan	Long	878 Market St	Greenville	MI	48838
18	141	Lily	Reed	666 State St	Albion	MI	49224
19	100	Liam	Anderson	101 Maple St	Detroit	MI	48201
20	138	Levi	Murphy	333 Depot St	Belding	MI	48809

Query executed successfully.

SQLQuery9....ADMIN (53))* | SQLQuery8.s...IADMIN (75))* | SQLQuery7.s...IADMIN (52))* | SQLQuery6.s...IADMIN (72))* | SQLQuery4.s

```sql
declare @num INT = 15;

IF @num > 10
    Print 'Greater than 10';
IF @num < 20
    Print 'Less than 20';
IF @num = 15
    Print 'Equals 15';
```

107 % No issues found

Messages

```
Greater than 10
Less than 20
Equals 15

Completion time: 2026-04-05T12:51:43.8657273-04:00
```

107 % No issues found

Query executed successfully.

Output

Show output from: Object Explorer

SQLQuery10...DMIN (76))* | SQLQuery9.s...IADMIN (53))* | SQLQuery8

```sql
DECLARE @score INT = 85;

IF @score >= 90
    PRINT 'A';
ELSE IF @score >= 80
    PRINT 'B';
ELSE
    PRINT 'C';
```

107 % | No issues found

Messages

```
B

Completion time: 2026-04-05T12:52:18.9432382-04:00
```

SQLQuery10...DMIN (76))* | SQLQuery9.s...IADMIN (53))* | SQLQuery8.

```sql
DECLARE @score INT = 90;

IF @score >= 90
    PRINT 'A';
ELSE IF @score >= 80
    PRINT 'B';
ELSE
    PRINT 'C';
```

107 % | No issues found

Messages

```
A

Completion time: 2026-04-05T12:52:51.7008038-04:00
```

SQLQuery10...DMIN (76))* | SQLQuery9.s...IADMIN (53))* | SQLQu

```sql
DECLARE @score INT = 5;

IF @score >= 90
    PRINT 'A';
ELSE IF @score >= 80
    PRINT 'B';
ELSE
    PRINT 'C';
```

107 % | No issues found

Messages

```
C

Completion time: 2026-04-05T12:53:18.7157172-04:00
```

the one logged into Windows. Depending on your configuration, you may experience a degraded sign-in experience for both Microsoft and GitHub accounts. Dism

SQLQuery11...DMIN (74))* | SQLQuery10....IADMIN (76))* | SQLQuery9.s...IADMIN (53))* | SQLQuery8.s...IADMIN (75))*

```sql
    CASE
        WHEN @grade >= 90 THEN 'A'
        WHEN @grade >= 80 THEN 'B'
        WHEN @grade >= 70 THEN 'C'
        ELSE 'F'
    END AS Grade;

    DECLARE @grade1 INT = 81;

    SELECT
    CASE
        WHEN @grade1 >= 90 THEN 'A'
        WHEN @grade1 >= 80 THEN 'B'
        WHEN @grade1 >= 70 THEN 'C'
        ELSE 'F'
    END AS Grade;

    DECLARE @grade2 INT = 91;

    SELECT
    CASE
        WHEN @grade2 >= 90 THEN 'A'
        WHEN @grade2 >= 80 THEN 'B'
        WHEN @grade2 >= 70 THEN 'C'
        ELSE 'F'
    END AS Grade;
```

97 % No issues found

Results | Messages

	Grade
1	C

	Grade
1	B

	Grade
1	A

Query executed successfully.

SQLQuery12...DMIN (81))* | SQLQuery11....IADMIN (74))* | SQLQuery10....IADMIN (76))* | SQLQu

```sql
ALTER TABLE Customer2
ADD LoyaltyPoints INT NULL;

Select * from Customer2
```

107 % 1 0

Results | Messages

	CustomerID	FirstName	LastName	DateOfBirth	Address	City	State	ZipCode	LoyaltyPoints
1	100	Liam	Anderson	1990-01-12	101 Maple St	Detroit	MI	48201	NULL
2	101	Olivia	Brown	1988-03-22	202 Oak St	Lansing	MI	48901	NULL
3	102	Noah	Clark	1992-05-14	303 Pine St	Flint	MI	48501	NULL
4	103	Emma	Davis	1995-07-30	404 Elm St	Grand Rapids	MI	49501	NULL
5	104	Oliver	Evans	1987-09-18	505 Cedar St	Ann Arbor	MI	48103	NULL
6	105	Ava	Garcia	1991-11-25	606 Birch St	Kalamazoo	MI	49001	NULL
7	106	Elijah	Harris	1989-02-10	707 Spruce St	Saginaw	MI	48601	NULL
8	107	Sophia	Johnson	1993-04-05	808 Willow St	Muskegon	MI	49440	NULL
9	108	James	King	1994-06-17	909 Cherry St	Midland	MI	48640	NULL
10	109	Isabella	Lewis	1990-08-29	111 Walnut St	Bay City	MI	48706	NULL
11	110	Benjamin	Martinez	1986-10-13	222 Poplar St	Traverse City	MI	49684	NULL
12	111	Mia	Moore	1992-12-01	333 Ash St	Jackson	MI	49201	NULL
13	112	Lucas	Nelson	1985-01-19	444 Beech St	Battle Creek	MI	49014	NULL
14	113	Charlotte	Ortiz	1993-03-27	555 Hickory St	Port Huron	MI	48060	NULL
15	114	Henry	Perez	1988-05-09	666 Dogwood St	Monroe	MI	48161	NULL
16	115	Amelia	Roberts	1991-07-21	777 Sycamore St	Mount Pleasant	MI	48858	NULL
17	116	Alexander	Scott	1994-09-11	888 Fir St	Ypsilanti	MI	48197	NULL
18	117	Harper	Turner	1987-11-03	999 Palm St	Adrian	MI	49221	NULL
19	118	Daniel	Walker	1990-02-14	121 Lake St	Marquette	MI	49855	NULL
20	119	Evelyn	Young	1995-04-26	232 Hill St	Escanaba	MI	49829	NULL

Tyrels-Laptop...bo.ErrorAudit | SQLQuery13...DMIN (77))* | SQLQuery12....IADMIN (81))* | SQLQuery11....IADMIN (74))*

```sql
Declare @ErrorHold INT = 0;
Insert into Customer2
    (CustomerID, FirstName)
Values
    (51, 'John')
Set @ErrorHold = @@ERROR
Print CONCAT('@@error is ', @ErrorHold)

If @ErrorHold != 0
    Insert into ErrorAudit
        (ErrorNbr)
        Values
            (@ErrorHold)

            --This is #12 using example from March 24th and 26th
```

107 % 2 0

Messages

```
Msg 544, Level 16, State 1, Line 2
Cannot insert explicit value for identity column in table 'Customer2' when IDENTITY_INSERT is set to OFF.
@@error is 544

(1 row affected)

Completion time: 2026-04-05T13:24:57.2515154-04:00
```

SQLQuery14...DMIN (84))* | Tyrels-Laptop...bo.ErrorAudit | SQLQuery13....IADMIN (77))* | SQLQuery12....IADMIN (81))* | SQLQuery11...

```sql
Declare @FName VARCHAR(50);
Set @FName = 'Emma';
--#13 of SQL 3 for ISIN 325 this is utilizing variables

Select *
From Customer2
Where Customer2.FirstName = @FName
```

172 % | No issues found

Results | Messages

	CustomerID	FirstName	LastName	DateOfBirth	Address	City	State	ZipCode	LoyaltyPoints
1	103	Emma	Davis	1995-07-30	404 Elm St	Grand Rapids	MI	49501	NULL

SQLQuery15...DMIN (87))* | SQLQuery14....IADMIN (84))* | Tyrels-Laptop...bo.ErrorAudit | SQ

```sql
CREATE PROCEDURE UpdateCustomerFirstName
    @CustID INT,
    @NewFirstName VARCHAR(50)
AS
BEGIN
    DECLARE @ErrorHold INT = 0;

    -- Attempt update
    UPDATE Customer2
    SET FirstName = @NewFirstName
    WHERE CustomerID = @CustID;

    -- Capture error
    SET @ErrorHold = @@ERROR;

    PRINT CONCAT('@@ERROR is ', @ErrorHold);

    -- Log error if it exists
    IF @ErrorHold != 0
    BEGIN
        INSERT INTO ErrorAudit (ErrorNbr)
        VALUES (@ErrorHold);
    END
END;

EXEC UpdateCustomerFirstName @CustID = 'One Hundred', @NewFirstName = 'Michael';
```

88 % | 2 | 0

Messages

```
Msg 8114, Level 16, State 5, Procedure UpdateCustomerFirstName, Line 0 [Batch Start Line 25]
Error converting data type varchar to int.

Completion time: 2026-04-05T13:38:44.1315923-04:00
```

SQLQuery19.s...lADMIN (88)) | SQLQuery15...DMIN (87))* | SQLQuery14....lADMIN (84))* | Tyrels-Laptop...bo.E

```sql
CREATE PROCEDURE UpdateCustomerFirstName1
    @CustID INT,
    @NewFirstName VARCHAR(50)
AS
BEGIN
   Declare @ErrorHold INT = 0;
    -- Attempt update
    UPDATE Customer2
    SET FirstName = @NewFirstName
    WHERE CustomerID = @CustID;

    -- Capture error
    SET @ErrorHold = @@ERROR;

    PRINT CONCAT('@@ERROR is ', @ErrorHold);

    -- Log error if it exists
    IF @ErrorHold != 0
    BEGIN
        INSERT INTO Meijer.dbo.ErrorAudit
        VALUES (@ErrorHold);
    END
END;

EXEC UpdateCustomerFirstName1 @CustID = NULL, @NewFirstName = 'Test';
```

97 % | 1 | 0

Messages

```
(0 rows affected)
@@ERROR is 0

(0 rows affected)
@@ERROR is 0

Completion time: 2026-04-05T14:10:15.1691777-04:00
```

SQLQuery20...DMIN (85))* | SQLQuery19.s...IADMIN (88)) | SQLQuery15....IADMIN (87))* | SQLQuery14....IADMIN (84))* | Ty

```sql
CREATE TABLE Cars (
    CarID INT IDENTITY(1,1),
    Make VARCHAR(50),
    Model VARCHAR(50),
    Year INT,
    Price DECIMAL(10,2),
    Mileage INT
);

DECLARE @i INT = 1;

WHILE @i <= 10
BEGIN
    INSERT INTO Cars (Make, Model, Year, Price, Mileage)
    VALUES ('Ford', 'Model' + CAST(@i AS VARCHAR), 2000 + @i, 20000 + (@i * 1000), @i * 5000);

    SET @i = @i + 1;
END;

Select * from Cars
```

97 % | 1 | 0

Results | Messages

	CarID	Make	Model	Year	Price	Mileage
1	1	Ford	Model1	2001	21000.00	5000
2	2	Ford	Model2	2002	22000.00	10000
3	3	Ford	Model3	2003	23000.00	15000
4	4	Ford	Model4	2004	24000.00	20000
5	5	Ford	Model5	2005	25000.00	25000
6	6	Ford	Model6	2006	26000.00	30000
7	7	Ford	Model7	2007	27000.00	35000
8	8	Ford	Model8	2008	28000.00	40000
9	9	Ford	Model9	2009	29000.00	45000
10	10	Ford	Model10	2010	30000.00	50000

SQLQuery21...DMIN (57))* | SQLQuery20....IADMIN (85))* | SQLQuery19.s...IADMIN (88)) | SQLQuery15....IADM

```sql
CREATE TABLE HikingLocations (
    LocationID INT,
    Name VARCHAR(100),
    State VARCHAR(50),
    CONSTRAINT PK_Hiking PRIMARY KEY (LocationID)
        WITH (IGNORE_DUP_KEY = ON)
);

INSERT INTO HikingLocations VALUES (1,'Trail A','MI');
INSERT INTO HikingLocations VALUES (2,'Trail b','MI');
INSERT INTO HikingLocations VALUES (1,'Trail A','MI'); -- ignored

Select * from HikingLocations
```

107 % | No issues found

Results | Messages

	LocationID	Name	State
1	1	Trail A	MI
2	2	Trail b	MI

SQLQuery22...DMIN (92))* | SQLQuery21....IADMIN (57))* | SQLQuery20....IADMIN (85))*

```sql
CREATE TABLE HikingLocations2 (
    LocationID INT,
    Name VARCHAR(100),
    State VARCHAR(50)
);

INSERT INTO HikingLocations2 VALUES
(1,'Trail A','MI'),
(2,'Trail B','MI'),
(3,'Trail C','MI'),
(1,'Trail A','MI'),
(2,'Trail B','MI'),
(4,'Trail D','MI'),
(5,'Trail E','MI'),
(6,'Trail F','MI'),
(3,'Trail C','MI'),
(7,'Trail G','MI');


SELECT LocationID, COUNT(*) AS CountOfDuplicates
FROM HikingLocations2
GROUP BY LocationID
HAVING COUNT(*) > 1;

SELECT *
FROM HikingLocations2
WHERE LocationID IN (
    SELECT LocationID
    FROM HikingLocations2
    GROUP BY LocationID
    HAVING COUNT(*) > 1
```

80 % No issues found

Results | Messages

	LocationID	CountOfDuplicates
1	1	2
2	2	2
3	3	2

	LocationID	Name	State
1	1	Trail A	MI
2	2	Trail B	MI
3	3	Trail C	MI
4	1	Trail A	MI
5	2	Trail B	MI
6	3	Trail C	MI

SQLQuery23...DMIN (59))* | SQLQuery22....IADMIN (92))* | SQLQuery21....IADI

```sql
DECLARE @ErrorHold INT = 0;          -- Variable to hold error number
DECLARE @LocationID INT;             -- Variable for LocationID
DECLARE @Name VARCHAR(100);          -- Variable for Name
DECLARE @State VARCHAR(50);          -- Variable for State

-- Example: Insert multiple records using variables
DECLARE @i INT = 1;

WHILE @i <= 10
BEGIN
    -- Assign values for each iteration
    SET @LocationID = @i;
    SET @Name = CONCAT('Trail ', CHAR(64 + @i));  -- Trail A, Trail B, ...
    SET @State = 'MI';

    -- Attempt insert
    INSERT INTO HikingLocations2 (LocationID, Name, State)
    VALUES (@LocationID, @Name, @State);

    -- Capture any error
    SET @ErrorHold = @@ERROR;
    PRINT CONCAT('Insert Attempt ', @i, ': @@ERROR = ', @ErrorHold);

    -- If error occurred, log it to ErrorAudit table
    IF @ErrorHold != 0
    BEGIN
        INSERT INTO ErrorAudit (ErrorNbr)
        VALUES (@ErrorHold);
    END

    -- Increment loop counter
    SET @i = @i + 1;
END
```

66 % 7 0

Messages

```
Insert Attempt 7: @@ERROR = 0

(1 row affected)
Insert Attempt 8: @@ERROR = 0

(1 row affected)
Insert Attempt 9: @@ERROR = 0

(1 row affected)
Insert Attempt 10: @@ERROR = 0

Completion time: 2026-04-05T14:55:28.4039375-04:00
```